Atiyah–Macdonald, Chapter 1: Rings and Ideals

Solutions to the exercises

Conventions. As in the book, "ring" means commutative ring with 1. $\mathfrak{N}$ is the nilradical, $\mathfrak{R}$ the Jacobson radical, $r(\mathfrak{a})$ the radical of an ideal, $X = \operatorname{Spec} A$. For a ring homomorphism $f : A \to B$, $\mathfrak{a}^{\mathrm{e}} = Bf(\mathfrak{a})$ and $\mathfrak{b}^{\mathrm{c}} = f^{-1}(\mathfrak{b})$.

Facts used freely from the text: (1.5) every non-unit lies in a maximal ideal; (1.7) $\mathfrak{N}$ is an ideal; (1.8) $\mathfrak{N} = \bigcap$ of all primes; (1.9) $x \in \mathfrak{R} \iff 1 - xy$ is a unit for all y; (1.14) $r(\mathfrak{a}) = \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \mathfrak{p}$.

This chapter covers the three in-text exercises (1.12, 1.13, 1.18) and the 28 end-of-chapter exercises. Exercise 27 contains no assignment as printed (it is exposition leading into Exercise 28); I comment on it at the appropriate place.

In-text exercises

Exercise 1.12

Problem. Let $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{a}_i, \mathfrak{b}_i$ be ideals of a ring A, and recall the *ideal quotient* $(\mathfrak{a} : \mathfrak{b}) = \{x \in A : x\mathfrak{b} \subseteq \mathfrak{a}\}$. Prove:

(i) $\mathfrak{a} \subseteq (\mathfrak{a} : \mathfrak{b})$;

(ii) $(\mathfrak{a} : \mathfrak{b})\mathfrak{b} \subseteq \mathfrak{a}$;

(iii) $((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c}) = (\mathfrak{a} : \mathfrak{b}\mathfrak{c}) = ((\mathfrak{a} : \mathfrak{c}) : \mathfrak{b})$;

(iv) $\big(\bigcap_i \mathfrak{a}_i : \mathfrak{b}\big) = \bigcap_i (\mathfrak{a}_i : \mathfrak{b})$;

(v) $\big(\mathfrak{a} : \sum_i \mathfrak{b}_i\big) = \bigcap_i (\mathfrak{a} : \mathfrak{b}_i)$.

Solution. (i) If $x \in \mathfrak{a}$ then $x\mathfrak{b} \subseteq \mathfrak{a}$ because $\mathfrak{a}$ is an ideal; so $\mathfrak{a} \subseteq (\mathfrak{a} : \mathfrak{b})$.

(ii) $(\mathfrak{a} : \mathfrak{b})\mathfrak{b}$ is generated by products xy with $x \in (\mathfrak{a} : \mathfrak{b})$, $y \in \mathfrak{b}$; each lies in $\mathfrak{a}$ by definition of $(\mathfrak{a} : \mathfrak{b})$.

(iii) For $x \in A$: $x \in ((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c}) \iff x\mathfrak{c} \subseteq (\mathfrak{a} : \mathfrak{b}) \iff xcb \in \mathfrak{a}$ for all $c \in \mathfrak{c}, b \in \mathfrak{b}$ $\iff x\,(\mathfrak{b}\mathfrak{c}) \subseteq \mathfrak{a}$ (the ideal $\mathfrak{b}\mathfrak{c}$ consists of finite sums of products bc, and $\mathfrak{a}$ is closed under sums) $\iff x \in (\mathfrak{a} : \mathfrak{b}\mathfrak{c})$. The condition "$xcb \in \mathfrak{a}$ for all b, c" is symmetric in $\mathfrak{b}, \mathfrak{c}$, so also $((\mathfrak{a} : \mathfrak{c}) : \mathfrak{b}) = (\mathfrak{a} : \mathfrak{b}\mathfrak{c})$.

(iv) $x\mathfrak{b} \subseteq \bigcap_i \mathfrak{a}_i \iff x\mathfrak{b} \subseteq \mathfrak{a}_i$ for every i.

(v) If $x\mathfrak{b}_i \subseteq \mathfrak{a}$ for all i, then $x\sum b_i \subseteq \mathfrak{a}$ (sums of elements of $\mathfrak{a}$). Conversely $\mathfrak{b}_i \subseteq \sum_j \mathfrak{b}_j$. Hence $(\mathfrak{a} : \sum_i \mathfrak{b}_i) = \bigcap_i (\mathfrak{a} : \mathfrak{b}_i)$. □

Exercise 1.13

Problem. Recall $r(\mathfrak{a}) = \{x \in A : x^n \in \mathfrak{a} \text{ for some } n > 0\}$. Prove:

(i) $r(\mathfrak{a}) \supseteq \mathfrak{a}$;
(ii) $r(r(\mathfrak{a})) = r(\mathfrak{a})$;
(iii) $r(\mathfrak{a}\mathfrak{b}) = r(\mathfrak{a} \cap \mathfrak{b}) = r(\mathfrak{a}) \cap r(\mathfrak{b})$;
(iv) $r(\mathfrak{a}) = (1) \iff \mathfrak{a} = (1)$;
(v) $r(\mathfrak{a} + \mathfrak{b}) = r(r(\mathfrak{a}) + r(\mathfrak{b}))$;
(vi) if $\mathfrak{p}$ is prime, $r(\mathfrak{p}^n) = \mathfrak{p}$ for all $n > 0$.

Solution. (i) $x \in \mathfrak{a} \Rightarrow x^1 \in \mathfrak{a}$.

(ii) $r(r(\mathfrak{a})) \supseteq r(\mathfrak{a})$ by (i). If $x^n \in r(\mathfrak{a})$ then $(x^n)^m \in \mathfrak{a}$ for some m, so $x \in r(\mathfrak{a})$.

(iii) Since $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b}$ we get $r(\mathfrak{a}\mathfrak{b}) \subseteq r(\mathfrak{a} \cap \mathfrak{b})$; since $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a}, \mathfrak{b}$ we get $r(\mathfrak{a} \cap \mathfrak{b}) \subseteq r(\mathfrak{a}) \cap r(\mathfrak{b})$. If $x \in r(\mathfrak{a}) \cap r(\mathfrak{b})$, say $x^m \in \mathfrak{a}$, $x^n \in \mathfrak{b}$, then $x^{m+n} \in \mathfrak{a}\mathfrak{b}$, so $x \in r(\mathfrak{a}\mathfrak{b})$. All three coincide.

(iv) $r(\mathfrak{a}) = (1) \iff 1 \in r(\mathfrak{a}) \iff 1^n \in \mathfrak{a} \iff \mathfrak{a} = (1)$.

(v) $\mathfrak{a} \subseteq r\mathfrak{a}$, so $\mathfrak{a} + \mathfrak{b} \subseteq r\mathfrak{a} + r\mathfrak{b}$, hence $r(\mathfrak{a} + \mathfrak{b}) \subseteq r(r\mathfrak{a} + r\mathfrak{b})$. Conversely $r\mathfrak{a}, r\mathfrak{b} \subseteq r(\mathfrak{a} + \mathfrak{b})$, so $r\mathfrak{a} + r\mathfrak{b} \subseteq r(\mathfrak{a} + \mathfrak{b})$, and applying r and (ii): $r(r\mathfrak{a} + r\mathfrak{b}) \subseteq r(r(\mathfrak{a} + \mathfrak{b})) = r(\mathfrak{a} + \mathfrak{b})$.

(vi) If $\mathfrak{p}$ is prime, $x^k \in \mathfrak{p} \Rightarrow x \in \mathfrak{p}$, so $r(\mathfrak{p}) = \mathfrak{p}$. Then $\mathfrak{p}^n \subseteq \mathfrak{p}$ gives $r(\mathfrak{p}^n) \subseteq \mathfrak{p}$; and $x \in \mathfrak{p} \Rightarrow x^n \in \mathfrak{p}^n$ gives $\mathfrak{p} \subseteq r(\mathfrak{p}^n)$. □

Exercise 1.18

Problem. Let $f : A \to B$ be a ring homomorphism, $\mathfrak{a}_1, \mathfrak{a}_2$ ideals of A and $\mathfrak{b}_1, \mathfrak{b}_2$ ideals of B. Then

$$\begin{array}{ll} (\mathfrak{a}_1 + \mathfrak{a}_2)^{\mathrm{e}} = \mathfrak{a}_1^{\mathrm{e}} + \mathfrak{a}_2^{\mathrm{e}}, & (\mathfrak{b}_1 + \mathfrak{b}_2)^{\mathrm{c}} \supseteq \mathfrak{b}_1^{\mathrm{c}} + \mathfrak{b}_2^{\mathrm{c}}, \\ (\mathfrak{a}_1 \cap \mathfrak{a}_2)^{\mathrm{e}} \subseteq \mathfrak{a}_1^{\mathrm{e}} \cap \mathfrak{a}_2^{\mathrm{e}}, & (\mathfrak{b}_1 \cap \mathfrak{b}_2)^{\mathrm{c}} = \mathfrak{b}_1^{\mathrm{c}} \cap \mathfrak{b}_2^{\mathrm{c}}, \\ (\mathfrak{a}_1\mathfrak{a}_2)^{\mathrm{e}} = \mathfrak{a}_1^{\mathrm{e}}\mathfrak{a}_2^{\mathrm{e}}, & (\mathfrak{b}_1\mathfrak{b}_2)^{\mathrm{c}} \supseteq \mathfrak{b}_1^{\mathrm{c}}\mathfrak{b}_2^{\mathrm{c}}, \\ (\mathfrak{a}_1 : \mathfrak{a}_2)^{\mathrm{e}} \subseteq (\mathfrak{a}_1^{\mathrm{e}} : \mathfrak{a}_2^{\mathrm{e}}), & (\mathfrak{b}_1 : \mathfrak{b}_2)^{\mathrm{c}} \subseteq (\mathfrak{b}_1^{\mathrm{c}} : \mathfrak{b}_2^{\mathrm{c}}), \\ r(\mathfrak{a})^{\mathrm{e}} \subseteq r(\mathfrak{a}^{\mathrm{e}}), & r(\mathfrak{b})^{\mathrm{c}} = r(\mathfrak{b}^{\mathrm{c}}). \end{array}$$

Deduce that the set E of extended ideals is closed under sum and product, and the set C of contracted ideals is closed under the other three operations ($\cap$, :, r).

Solution. Let $f : A \to B$.

Extensions.

- $(\mathfrak{a}_1 + \mathfrak{a}_2)^{\mathrm{e}} = \mathfrak{a}_1^{\mathrm{e}} + \mathfrak{a}_2^{\mathrm{e}}$: $\supseteq$ since $\mathfrak{a}_i \subseteq \mathfrak{a}_1 + \mathfrak{a}_2$; $\subseteq$ since the generators $f(x_1 + x_2) = f(x_1) + f(x_2)$ lie in $\mathfrak{a}_1^{\mathrm{e}} + \mathfrak{a}_2^{\mathrm{e}}$.
- $(\mathfrak{a}_1 \cap \mathfrak{a}_2)^{\mathrm{e}} \subseteq \mathfrak{a}_1^{\mathrm{e}} \cap \mathfrak{a}_2^{\mathrm{e}}$ since $\mathfrak{a}_1 \cap \mathfrak{a}_2 \subseteq \mathfrak{a}_i$.

- $(\mathfrak{a}_1\mathfrak{a}_2)^{\mathrm{e}} = \mathfrak{a}_1^{\mathrm{e}}\mathfrak{a}_2^{\mathrm{e}}$: the generators $f(xy) = f(x)f(y)$ of the left side lie in the right; conversely the right side is generated by products $\big(\sum b_i f(x_i)\big)\big(\sum b'_j f(x'_j)\big) = \sum b_i b'_j f(x_i x'_j) \in (\mathfrak{a}_1\mathfrak{a}_2)^{\mathrm{e}}$.
- $(\mathfrak{a}_1 : \mathfrak{a}_2)^{\mathrm{e}} \subseteq (\mathfrak{a}_1^{\mathrm{e}} : \mathfrak{a}_2^{\mathrm{e}})$: if $x \in (\mathfrak{a}_1 : \mathfrak{a}_2)$ then $f(x)f(y) = f(xy) \in f(\mathfrak{a}_1)$ for $y \in \mathfrak{a}_2$, so $f(x)\mathfrak{a}_2^{\mathrm{e}} \subseteq \mathfrak{a}_1^{\mathrm{e}}$, i.e. $f(x) \in (\mathfrak{a}_1^{\mathrm{e}} : \mathfrak{a}_2^{\mathrm{e}})$; the latter is an ideal, hence contains the ideal generated by these $f(x)$.
- $r(\mathfrak{a})^{\mathrm{e}} \subseteq r(\mathfrak{a}^{\mathrm{e}})$: if $x^n \in \mathfrak{a}$ then $f(x)^n \in \mathfrak{a}^{\mathrm{e}}$, so $f(x) \in r(\mathfrak{a}^{\mathrm{e}})$, an ideal.

Contractions. Preimages respect all inclusions, so:

- $(\mathfrak{b}_1 + \mathfrak{b}_2)^{\mathrm{c}} \supseteq \mathfrak{b}_1^{\mathrm{c}} + \mathfrak{b}_2^{\mathrm{c}}$ (monotonicity).
- $(\mathfrak{b}_1 \cap \mathfrak{b}_2)^{\mathrm{c}} = \mathfrak{b}_1^{\mathrm{c}} \cap \mathfrak{b}_2^{\mathrm{c}}$ (preimage of an intersection).
- $(\mathfrak{b}_1\mathfrak{b}_2)^{\mathrm{c}} \supseteq \mathfrak{b}_1^{\mathrm{c}}\mathfrak{b}_2^{\mathrm{c}}$: for $x \in \mathfrak{b}_1^{\mathrm{c}}, y \in \mathfrak{b}_2^{\mathrm{c}}$, $f(xy) = f(x)f(y) \in \mathfrak{b}_1\mathfrak{b}_2$.
- $(\mathfrak{b}_1 : \mathfrak{b}_2)^{\mathrm{c}} \subseteq (\mathfrak{b}_1^{\mathrm{c}} : \mathfrak{b}_2^{\mathrm{c}})$: if $f(x)\mathfrak{b}_2 \subseteq \mathfrak{b}_1$ and $y \in \mathfrak{b}_2^{\mathrm{c}}$ then $f(xy) = f(x)f(y) \in \mathfrak{b}_1$, so $xy \in \mathfrak{b}_1^{\mathrm{c}}$.
- $r(\mathfrak{b})^{\mathrm{c}} = r(\mathfrak{b}^{\mathrm{c}})$: $x \in r(\mathfrak{b})^{\mathrm{c}} \iff f(x)^n \in \mathfrak{b}$ some $n \iff x^n \in \mathfrak{b}^{\mathrm{c}}$ some n.

E is closed under $+, \cdot$; C is closed under $\cap, :, r$. Recall $C = \{\mathfrak{a} : \mathfrak{a} = \mathfrak{a}^{\mathrm{ec}}\}$, $E = \{\mathfrak{b} : \mathfrak{b} = \mathfrak{b}^{\mathrm{ce}}\}$.

If $\mathfrak{b}_1, \mathfrak{b}_2 \in E$, then $\mathfrak{b}_1 + \mathfrak{b}_2 = \mathfrak{b}_1^{\mathrm{ce}} + \mathfrak{b}_2^{\mathrm{ce}} = (\mathfrak{b}_1^{\mathrm{c}} + \mathfrak{b}_2^{\mathrm{c}})^{\mathrm{e}} \in E$ and $\mathfrak{b}_1\mathfrak{b}_2 = (\mathfrak{b}_1^{\mathrm{c}})^{\mathrm{e}}(\mathfrak{b}_2^{\mathrm{c}})^{\mathrm{e}} = (\mathfrak{b}_1^{\mathrm{c}}\mathfrak{b}_2^{\mathrm{c}})^{\mathrm{e}} \in E$ (extended ideals are in E by 1.17).

If $\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a} \in C$ then, using the displayed inclusions and $\mathfrak{a} = \mathfrak{a}^{\mathrm{ec}}$:

$$\begin{aligned}(\mathfrak{a}_1 \cap \mathfrak{a}_2)^{\mathrm{ec}} &\subseteq (\mathfrak{a}_1^{\mathrm{e}} \cap \mathfrak{a}_2^{\mathrm{e}})^{\mathrm{c}} = \mathfrak{a}_1^{\mathrm{ec}} \cap \mathfrak{a}_2^{\mathrm{ec}} = \mathfrak{a}_1 \cap \mathfrak{a}_2,\\ (\mathfrak{a}_1 : \mathfrak{a}_2)^{\mathrm{ec}} &\subseteq (\mathfrak{a}_1^{\mathrm{e}} : \mathfrak{a}_2^{\mathrm{e}})^{\mathrm{c}} \subseteq (\mathfrak{a}_1^{\mathrm{ec}} : \mathfrak{a}_2^{\mathrm{ec}}) = (\mathfrak{a}_1 : \mathfrak{a}_2),\\ r(\mathfrak{a})^{\mathrm{ec}} &\subseteq r(\mathfrak{a}^{\mathrm{e}})^{\mathrm{c}} = r(\mathfrak{a}^{\mathrm{ec}}) = r(\mathfrak{a}).\end{aligned}$$

The reverse inclusions hold always (1.17(i)), so these ideals lie in C. □

End-of-chapter exercises

Exercise 1

> **Problem.** Let x be a nilpotent element of a ring A. Show that $1 + x$ is a unit of A. Deduce that the sum of a nilpotent element and a unit is a unit.

Solution. Let $x^n = 0$. Then $(1 + x)\big(1 - x + x^2 - \cdots + (-1)^{n-1}x^{n-1}\big) = 1 + (-1)^{n-1}x^n = 1$. If u is a unit and x nilpotent, then $u^{-1}x$ is nilpotent (commutativity) so $u + x = u(1 + u^{-1}x)$ is a product of units. □

Exercise 2

> **Problem.** Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x, with coefficients in A. Let $f = a_0 + a_1x + \cdots + a_nx^n \in A[x]$. Prove that

(i) f is a unit in $A[x] \iff a_0$ is a unit in A and $a_1, \ldots, a_n$ are nilpotent. [If $b_0 + b_1x + \cdots + b_mx^m$ is the inverse of f, prove by induction on r that $a_n^{r+1}b_{m-r} = 0$. Hence show that a_n is nilpotent, and then use Ex. 1.]

(ii) f is nilpotent $\iff a_0, a_1, \ldots, a_n$ are nilpotent.

(iii) f is a zero-divisor $\iff$ there exists $a \neq 0$ in A such that $af = 0$. [Choose a polynomial $g = b_0 + b_1x + \cdots + b_mx^m$ of least degree m such that $fg = 0$. Then $a_nb_m = 0$, hence $a_ng = 0$ (because a_ng annihilates f and has degree $< m$). Now show by induction that $a_{n-r}g = 0$ $(0 \leq r \leq n)$.]

(iv) f is said to be *primitive* if $(a_0, a_1, \ldots, a_n) = (1)$. Prove that if $f, g \in A[x]$, then fg is primitive $\iff f$ and g are primitive.

Solution. Let $f = a_0 + a_1x + \cdots + a_nx^n$.

(i) f is a unit $\iff a_0$ is a unit and $a_1, \ldots, a_n$ are nilpotent.

($\Leftarrow$) Each a_ix^i $(i \geq 1)$ is nilpotent, and nilpotents form an ideal (1.7), so $f - a_0$ is nilpotent; apply Exercise 1.

($\Rightarrow$) Let $fg = 1$, $g = b_0 + \cdots + b_mx^m$. Then $a_0b_0 = 1$, so a_0 is a unit. We prove by induction on r $(0 \leq r \leq m)$ that

$$S(r): \quad a_n^{r+1}b_{m-r} = 0.$$

Assume $n \geq 1$ (else nothing to prove). $S(0)$: the coefficient of x^{n+m} in fg is $a_nb_m = 0$. For $r \geq 1$ assume $S(0), \ldots, S(r-1)$. The coefficient of x^{n+m-r} $(\geq n \geq 1)$ in fg is zero: $\sum_{k=0}^{r} a_{n-k}b_{m-r+k} = 0$ (with $a_i = 0$ for $i < 0$). Multiply by a_n^r:

$$a_n^{r+1}b_{m-r} + \sum_{k=1}^{r} a_{n-k}\, a_n^{k-1}\big(a_n^{r-k+1}b_{m-(r-k)}\big) = 0,$$

and each bracket vanishes by $S(r-k)$. So $S(r)$ holds. Taking $r = m$: $a_n^{m+1}b_0 = 0$, and b_0 is a unit, so a_n is nilpotent.

Now a_nx^n is nilpotent, so $f - a_nx^n$ is a unit (unit minus nilpotent, Ex. 1) of degree $< n$, with the same $a_0, \ldots, a_{n-1}$. By induction on n (the case $n = 0$ is trivial) $a_1, \ldots, a_{n-1}$ are nilpotent.

(ii) f nilpotent $\iff$ all a_i nilpotent. ($\Leftarrow$) as in (i): $\mathfrak{N}(A[x])$ is an ideal containing each a_ix^i. ($\Rightarrow$) Let $\mathfrak{p}$ be a prime of A. Then $\mathfrak{p}A[x]$ (polynomials with all coefficients in $\mathfrak{p}$) is prime in $A[x]$, because $A[x]/\mathfrak{p}A[x] \cong (A/\mathfrak{p})[x]$ is a domain. A nilpotent f lies in every prime of $A[x]$ (1.8), so all $a_i \in \mathfrak{p}$. Hence $a_i \in \bigcap \mathfrak{p} = \mathfrak{N}(A)$.

(iii) f is a zero-divisor $\iff \exists a \neq 0$ with $af = 0$. ($\Leftarrow$) clear. ($\Rightarrow$) Choose $g = b_0 + \cdots + b_mx^m \neq 0$ with $fg = 0$ and m minimal. Then $a_nb_m = 0$, so a_ng has degree $< m$ and $f(a_ng) = 0$; minimality gives $a_ng = 0$. We show by induction on $r = 0, \ldots, n$ that $a_{n-r}g = 0$. Suppose $a_{n-s}g = 0$ for $s < r$. The coefficient of x^{n+m-r} in fg is $\sum_{s=0}^{r} a_{n-s}b_{m-r+s} = 0$; the terms with $s < r$ vanish, so $a_{n-r}b_m = 0$. Then $a_{n-r}g$ has degree $< m$ and is killed by f, so $a_{n-r}g = 0$. Hence $a_ig = 0$ for all i; in particular $a := b_m \neq 0$ satisfies $a_ib_m = 0$ for all i, i.e. $b_mf = 0$.

(iv) For $f, g \in A[x]$: fg primitive $\iff$ f and g primitive. Let $c(f)$ be the ideal generated by the coefficients. Each coefficient of fg is a sum of products a_ib_j, so $c(fg) \subseteq c(f) \cap c(g)$; hence fg primitive forces $c(f) = c(g) = (1)$. Conversely suppose f, g primitive but $c(fg) \neq (1)$. Then $c(fg) \subseteq \mathfrak{m}$ for a maximal ideal $\mathfrak{m}$, so $fg \in \mathfrak{m}A[x]$, which is prime ($A[x]/\mathfrak{m}A[x] = (A/\mathfrak{m})[x]$ is a domain). Thus $f \in \mathfrak{m}A[x]$ or $g \in \mathfrak{m}A[x]$, i.e. $c(f) \subseteq \mathfrak{m}$ or $c(g) \subseteq \mathfrak{m}$, contradicting primitivity. $\square$

Exercise 3

Problem. Generalize the results of Exercise 2 to a polynomial ring $A[x_1, \ldots, x_r]$ in several indeterminates.

Solution. Write $A_r = A[x_1, \ldots, x_r]$. The statements are:

(i) $f \in A_r$ is a unit $\iff$ its constant term is a unit of A and all its other coefficients are nilpotent;

(ii) f is nilpotent $\iff$ all its coefficients are nilpotent;

(iii) f is a zero-divisor $\iff \exists a \in A$, $a \neq 0$, $af = 0$;

(iv) f, g are primitive (coefficients generate (1)) $\iff$ fg is primitive.

(i), (ii): induct on r, using $A_r = A_{r-1}[x_r]$. Write $f = \sum_k f_k x_r^k$ with $f_k \in A_{r-1}$. By Ex. 2, f is a unit iff f_0 is a unit and f_k ($k \geq 1$) are nilpotent in A_{r-1}; by induction this says: the constant term of f_0 (= constant term of f) is a unit and all other coefficients of f_0 are nilpotent, and all coefficients of the f_k, $k \geq 1$, are nilpotent. Same for (ii).

(iii) ($\Leftarrow$) clear. ($\Rightarrow$) (Kronecker substitution.) Let $fg = 0$, $g \neq 0$. Choose d larger than every exponent of every variable occurring in f or g, and let $\kappa : A_r \to A[t]$ be the ring homomorphism $x_i \mapsto t^{d^{i-1}}$. By uniqueness of base-d expansions, distinct monomials with all exponents $< d$ go to distinct powers of t; so κ is injective on the polynomials all of whose exponents are $< d$ (this includes f, g, and af for $a \in A$). Thus $\kappa(g) \neq 0$ and $\kappa(f)\kappa(g) = 0$, so $\kappa(f)$ is a zero-divisor of $A[t]$, and by Ex. 2(iii) there is $a \neq 0$ with $a\kappa(f) = \kappa(af) = 0$; injectivity gives $af = 0$.

(iv) Identical to Ex. 2(iv): $c(fg) \subseteq c(f) \cap c(g)$ for $\Rightarrow$, and for $\Leftarrow$ use $\mathfrak{m}A_r$ prime since $A_r/\mathfrak{m}A_r \cong (A/\mathfrak{m})[x_1, \ldots, x_r]$ is a domain. □

Exercise 4

Problem. In the ring $A[x]$, the Jacobson radical is equal to the nilradical.

Solution. Always $\mathfrak{N} \subseteq \mathfrak{R}$ (nilpotent elements lie in every prime, in particular every maximal ideal). Let $f = \sum a_i x^i \in \mathfrak{R}(A[x])$. By 1.9 (with $y = x$), $1 - xf = 1 - a_0 x - a_1 x^2 - \cdots$ is a unit. By Ex. 2(i), all of $-a_0, -a_1, \ldots$ are nilpotent. By Ex. 2(ii), f is nilpotent. □

Exercise 5

Problem. Let A be a ring and let $A[[x]]$ be the ring of formal power series $f = \sum_{n=0}^{\infty} a_n x^n$ with coefficients in A. Show that

(i) f is a unit in $A[[x]] \iff a_0$ is a unit in A.

(ii) If f is nilpotent, then a_n is nilpotent for all $n \geq 0$. Is the converse true? (See Chapter 7, Exercise 2.)

(iii) f belongs to the Jacobson radical of $A[[x]] \iff a_0$ belongs to the Jacobson radical of A.

(iv) The contraction of a maximal ideal $\mathfrak{m}$ of $A[[x]]$ is a maximal ideal of A, and $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x.

(v) Every prime ideal of A is the contraction of a prime ideal of $A[[x]]$.

Solution. Let $f = \sum a_n x^n$.

(i) If $fg = 1$ then $a_0 b_0 = 1$. Conversely if a_0 is a unit, define $b_0 = a_0^{-1}$ and recursively $b_n = -a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i}$; then $g = \sum b_n x^n$ satisfies $fg = 1$.

(ii) Let $f^N = 0$ and let $\mathfrak{p}$ be a prime of A. Reducing coefficients gives a ring homomorphism $A[[x]] \to (A/\mathfrak{p})[[x]]$, and $(A/\mathfrak{p})[[x]]$ is a domain (the product of the lowest-order terms $\bar{a}_r x^r, \bar{b}_s x^s$ of two nonzero series has coefficient $\bar{a}_r \bar{b}_s \neq 0$ at x^{r+s}). So $\bar{f} = 0$, i.e. all $a_n \in \mathfrak{p}$. Hence $a_n \in \bigcap \mathfrak{p} = \mathfrak{N}(A)$.

The converse is false. Let k be a field and $A = k[t_1, t_2, \dots]/(t_1, t_2^2, t_3^3, \dots)$. Each t_n is nilpotent ($t_n^n = 0$). Put $f = \sum_{n \geq 1} t_n x^n$. Suppose $f^N = 0$ and take $n > N$. The coefficient of x^{nN} in f^N is $\sum t_{i_1} \cdots t_{i_N}$ over ordered tuples with $i_1 + \cdots + i_N = nN$. A has the k-basis of monomials $\prod t_i^{e_i}$ with $e_i < i$; the monomial t_n^N (note $N < n$) is such a basis element, and it arises from only one tuple, $(n, \dots, n)$, while every other tuple gives a different monomial (or 0). So this coefficient is nonzero, contradicting $f^N = 0$. So f is not nilpotent although all its coefficients are.

(iii) By 1.9, $f \in \mathfrak{R}(A[[x]]) \iff 1 - fg$ is a unit for all $g \in A[[x]] \iff$ (by (i)) $1 - a_0 b_0$ is a unit in A for all g, i.e. for all $b_0 \in A$ (constants are allowed as g) $\iff a_0 \in \mathfrak{R}(A)$.

(iv) Let $\mathfrak{m}$ be maximal in $A[[x]]$. First $x \in \mathfrak{m}$: otherwise $\mathfrak{m} + (x) = (1)$, so $1 = h + gx$ with $h \in \mathfrak{m}$, and $h = 1 - gx$ is a unit by (i), absurd. Let $\varepsilon : A[[x]] \to A$, $f \mapsto a_0$, with kernel (x); it is surjective. Since $(x) \subseteq \mathfrak{m}$, $\mathfrak{m} = \varepsilon^{-1}(\mathfrak{m}')$ with $\mathfrak{m}' = \varepsilon(\mathfrak{m})$ maximal in A (Prop. 1.1: $\mathfrak{m}/(x)$ is maximal in $A[[x]]/(x) \cong A$). Constants satisfy $\varepsilon(a) = a$, so $\mathfrak{m}^c = \mathfrak{m} \cap A = \mathfrak{m}'$ is maximal in A. Finally every $f \in \mathfrak{m}$ is $a_0 + xh$ with $a_0 \in \mathfrak{m}' = \mathfrak{m}^c$, so $\mathfrak{m} = \mathfrak{m}^e + (x)$, i.e. $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x; the reverse inclusion is obvious.

(v) For a prime $\mathfrak{p}$ of A let $P = \varepsilon^{-1}(\mathfrak{p}) = \{f : a_0 \in \mathfrak{p}\}$. Then $A[[x]]/P \cong A/\mathfrak{p}$ is a domain, so P is prime, and $P^c = P \cap A = \mathfrak{p}$. □

Exercise 6

Problem. A ring A is such that every ideal not contained in the nilradical contains a nonzero idempotent (that is, an element e such that $e^2 = e \neq 0$). Prove that the nilradical and Jacobson radical of A are equal.

Solution. $\mathfrak{N} \subseteq \mathfrak{R}$ always. Suppose $\mathfrak{R} \not\subseteq \mathfrak{N}$. Then the ideal $\mathfrak{R}$ contains an idempotent $e \neq 0$. By 1.9, $1 - e = 1 - e \cdot 1$ is a unit. But $e(1 - e) = e - e^2 = 0$, so $e = 0$, contradiction. Hence $\mathfrak{R} \subseteq \mathfrak{N}$. □

Exercise 7

Problem. Let A be a ring in which every element x satisfies $x^n = x$ for some $n > 1$ (depending on x). Show that every prime ideal in A is maximal.

Solution. Let $\mathfrak{p}$ be prime and $D = A/\mathfrak{p}$, a domain in which every element also satisfies $\bar{x}^n = \bar{x}$, some $n > 1$. If $\bar{x} \neq 0$ then $\bar{x}(\bar{x}^{n-1} - 1) = 0$ forces $\bar{x}^{n-1} = 1$, i.e. $\bar{x} \cdot \bar{x}^{n-2} = 1$; so $\bar{x}$ is a unit. Thus D is a field and $\mathfrak{p}$ is maximal. □

Exercise 8

> **Problem.** Let A be a ring $\neq 0$. Show that the set of prime ideals of A has minimal elements with respect to inclusion.

Solution. Let Σ be the set of primes, ordered by reverse inclusion; it is nonempty since $A \neq 0$ has a maximal ideal. We show every chain $(\mathfrak{p}_\alpha)$ has an upper bound (i.e. the intersection $\mathfrak{p} = \bigcap \mathfrak{p}_\alpha$ is prime). $1 \notin \mathfrak{p}$. Suppose $xy \in \mathfrak{p}$ and $x \notin \mathfrak{p}$, say $x \notin \mathfrak{p}_\beta$. Let α be arbitrary. If $\mathfrak{p}_\alpha \subseteq \mathfrak{p}_\beta$, then $x \notin \mathfrak{p}_\alpha$ and $xy \in \mathfrak{p}_\alpha$ give $y \in \mathfrak{p}_\alpha$. If $\mathfrak{p}_\beta \subseteq \mathfrak{p}_\alpha$, then $xy \in \mathfrak{p}_\beta$ and $x \notin \mathfrak{p}_\beta$ give $y \in \mathfrak{p}_\beta \subseteq \mathfrak{p}_\alpha$. So $y \in \mathfrak{p}$. By Zorn, Σ has a maximal element for reverse inclusion, i.e. a minimal prime ideal. (Indeed every prime contains a minimal prime.) □

Exercise 9

> **Problem.** Let $\mathfrak{a}$ be an ideal $\neq (1)$ in a ring A. Show that $\mathfrak{a} = r(\mathfrak{a}) \iff \mathfrak{a}$ is an intersection of prime ideals.

Solution. ($\Rightarrow$) $r(\mathfrak{a}) = \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \mathfrak{p}$ by 1.14. ($\Leftarrow$) If $\mathfrak{a} = \bigcap_i \mathfrak{p}_i$ then $r(\mathfrak{a}) \subseteq r(\mathfrak{p}_i) = \mathfrak{p}_i$ for each i, so $r(\mathfrak{a}) \subseteq \mathfrak{a}$; and $\mathfrak{a} \subseteq r(\mathfrak{a})$ always. □

Exercise 10

> **Problem.** Let A be a ring, $\mathfrak{N}$ its nilradical. Show that the following are equivalent:
>
> (i) A has exactly one prime ideal;
>
> (ii) every element of A is either a unit or nilpotent;
>
> (iii) $A/\mathfrak{N}$ is a field.

Solution. (Assume $A \neq 0$.) (i)$\Rightarrow$(ii): If $\mathfrak{p}$ is the only prime then $\mathfrak{N} = \mathfrak{p}$ (1.8). A non-unit lies in a maximal ideal, which is prime, hence equals $\mathfrak{p} = \mathfrak{N}$, so the non-unit is nilpotent. (ii)$\Rightarrow$(iii): $A/\mathfrak{N} \neq 0$. If $\bar{x} \neq 0$ then x is not nilpotent, hence a unit, hence $\bar{x}$ is a unit. (iii)$\Rightarrow$(i): $\mathfrak{N}$ is maximal, hence prime. Every prime contains $\mathfrak{N} = \bigcap \mathfrak{p}$, hence equals the maximal ideal $\mathfrak{N}$. □

Exercise 11

> **Problem.** A ring A is *Boolean* if $x^2 = x$ for all $x \in A$. In a Boolean ring A, show that
>
> (i) $2x = 0$ for all $x \in A$;
>
> (ii) every prime ideal $\mathfrak{p}$ is maximal, and $A/\mathfrak{p}$ is a field with two elements;
>
> (iii) every finitely generated ideal in A is principal.

Solution. (i) Apply $y^2 = y$ to $y = 2x$: $2x = (2x)^2 = 4x^2 = 4x$, hence $2x = 4x - 2x = 0$.

(ii) Let $\mathfrak{p}$ be prime. In the domain $A/\mathfrak{p}$ every $\bar{x}$ satisfies $\bar{x}(\bar{x} - 1) = 0$, so $\bar{x} \in \{0, 1\}$. Thus $A/\mathfrak{p} = \{0, 1\} \cong \mathbf{F}_2$ is a field and $\mathfrak{p}$ is maximal.

(iii) For $x, y \in A$ put $z = x + y + xy \in (x, y)$. Then $xz = x + xy + xy = x + 2xy = x$ and likewise $yz = y$, so $(x, y) \subseteq (z) \subseteq (x, y)$. By induction on the number of generators every finitely generated ideal is principal. □

Exercise 12

Problem. A local ring contains no idempotent $\neq 0, 1$.

Solution. Let $\mathfrak{m}$ be the maximal ideal and $e^2 = e$, so $e(1 - e) = 0$. Not both $e, 1 - e$ lie in $\mathfrak{m}$ (else $1 \in \mathfrak{m}$). If $e \notin \mathfrak{m}$, e is a unit (all elements outside $\mathfrak{m}$ are, by 1.6), so $1 - e = 0$, $e = 1$. If $1 - e \notin \mathfrak{m}$, then $1 - e$ is a unit and $e = 0$. □

Exercise 13

Problem. (*Construction of an algebraic closure of a field*, E. Artin.) Let K be a field and let Σ be the set of all irreducible monic polynomials f in one indeterminate with coefficients in K. Let A be the polynomial ring over K generated by indeterminates x_f, one for each $f \in \Sigma$. Let $\mathfrak{a}$ be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. Show that $\mathfrak{a} \neq (1)$.

Let $\mathfrak{m}$ be a maximal ideal of A containing $\mathfrak{a}$, and let $K_1 = A/\mathfrak{m}$. Then K_1 is an extension field of K in which each $f \in \Sigma$ has a root. Repeat the construction with K_1 in place of K, obtaining a field K_2, and so on. Let $L = \bigcup_{n=1}^{\infty} K_n$. Then L is a field in which each $f \in \Sigma$ splits completely into linear factors. Let $\bar{K}$ be the set of all elements of L which are algebraic over K. Then $\bar{K}$ is an algebraic closure of K.

Solution. Let $A = K[x_f : f \in \Sigma]$ and $\mathfrak{a} = \big(f(x_f) : f \in \Sigma\big)$.

$\mathfrak{a} \neq (1)$. Suppose $1 = \sum_{i=1}^{n} g_i f_i(x_{f_i})$ with $g_i \in A$ and $f_1, \ldots, f_n \in \Sigma$ distinct. Let F be a finite extension field of K in which each f_i has a root α_i (adjoin roots one at a time: $K[t]/$(irreducible factor)). The K-algebra homomorphism $A \to F$ with $x_{f_i} \mapsto \alpha_i$ $(i \leq n)$ and all other $x_f \mapsto 0$ sends the relation to $1 = 0$, absurd.

The tower. Take a maximal ideal $\mathfrak{m} \supseteq \mathfrak{a}$ (1.4) and $K_1 = A/\mathfrak{m}$. Since K is a field, $K \to K_1$ is injective; the image of x_f is a root of f in K_1. Repeat with K_1 in place of K to get K_2, etc., and put $L = \bigcup_n K_n$, a field (union of a chain of fields).

Claim: every non-constant $h \in K_n[t]$ has a root in K_{n+1}. Let g be a monic irreducible factor of h over K_n; g has a root in K_{n+1} by construction, and it is a root of h. Hence if $f \in K[t]$ has degree d, it has a root in K_1, so $f = (t - \alpha_1) f_1$ in $K_1[t]$; f_1 has a root in K_2; etc. After d steps f splits into linear factors in $K_d[t] \subseteq L[t]$. So every $f \in \Sigma$ splits completely in L.

$\bar{K}$ is an algebraic closure. Let $\bar{K} \subseteq L$ be the set of elements algebraic over K; it is a subfield (standard: $K(\alpha, \beta)$ is finite over K). Every $f \in K[t]$ splits into linear factors in $L[t]$, and its roots there are algebraic over K, so f splits completely in $\bar{K}[t]$. Now let $h \in \bar{K}[t]$ be irreducible

and let β be a root of h in the extension field $E = \bar{K}[t]/(h)$. Then β is algebraic over $\bar{K}$, which is algebraic over K, so β is algebraic over K (finite towers), i.e. a root of some $f \in K[t]$, $f \neq 0$. As $f = \prod(t - \gamma_i)$ with $\gamma_i \in \bar{K}$, $\beta = \gamma_i \in \bar{K}$. So h has a root in $\bar{K}$, hence is linear. Thus $\bar{K}$ is algebraically closed and algebraic over K, i.e. an algebraic closure of K. □

Exercise 14

> **Problem.** In a ring A, let Σ be the set of all ideals in which every element is a zero-divisor. Show that the set Σ has maximal elements and that every maximal element of Σ is a prime ideal. Hence the set of zero-divisors in A is a union of prime ideals.

Solution. (Assume $A \neq 0$.) $\Sigma \ni (0)$ is nonempty. The union of a chain in Σ is an ideal all of whose elements are zero-divisors; so Zorn gives maximal elements of Σ, and more precisely, for each ideal $\mathfrak{a} \in \Sigma$, a maximal element of Σ containing it.

Let $\mathfrak{p}$ be maximal in Σ. $\mathfrak{p} \neq (1)$ since 1 is not a zero-divisor. Suppose $xy \in \mathfrak{p}$ with $x, y \notin \mathfrak{p}$. By maximality $\mathfrak{p} + (x) \notin \Sigma$ and $\mathfrak{p} + (y) \notin \Sigma$, so there exist non-zero-divisors $u = p_1 + ax$ and $v = p_2 + by$ ($p_i \in \mathfrak{p}$). Then $uv = p_1p_2 + p_1by + p_2ax + ab\,xy \in \mathfrak{p}$, and uv is a non-zero-divisor (a product of two non-zero-divisors), contradicting $\mathfrak{p} \in \Sigma$. So $\mathfrak{p}$ is prime.

Finally let D be the set of zero-divisors. Each maximal element of Σ is a prime contained in D. Conversely, if $x \in D$ with $xy = 0$, $y \neq 0$, then every ax is a zero-divisor ($axy = 0$), so $(x) \in \Sigma$ and (x) lies in a maximal element of Σ, a prime. Therefore D is the union of these primes. □

Exercise 15

> **Problem.** (*The prime spectrum of a ring.*) Let A be a ring and let X be the set of all prime ideals of A. For each subset E of A, let $V(E)$ denote the set of all prime ideals of A which contain E. Prove that
>
> (i) if $\mathfrak{a}$ is the ideal generated by E, then $V(E) = V(\mathfrak{a}) = V(r(\mathfrak{a}))$;
> (ii) $V(0) = X$, $V(1) = \emptyset$;
> (iii) if $(E_i)_{i \in I}$ is any family of subsets of A, then $V\big(\bigcup_{i \in I} E_i\big) = \bigcap_{i \in I} V(E_i)$;
> (iv) $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$ for any ideals $\mathfrak{a}, \mathfrak{b}$ of A.
>
> These results show that the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the *Zariski topology*. The topological space X is called the *prime spectrum* of A, and is written $\mathrm{Spec}(A)$.

Solution. (i) A prime contains $E \iff$ it contains the ideal $\mathfrak{a} = (E)$ generated by E. A prime $\mathfrak{p} \supseteq \mathfrak{a}$ contains $r(\mathfrak{a})$ (if $x^n \in \mathfrak{a} \subseteq \mathfrak{p}$ then $x \in \mathfrak{p}$); conversely $\mathfrak{p} \supseteq r(\mathfrak{a}) \supseteq \mathfrak{a}$. So $V(E) = V(\mathfrak{a}) = V(r\mathfrak{a})$.

(ii) Every prime contains 0: $V(0) = X$. No prime contains 1: $V(1) = \emptyset$.

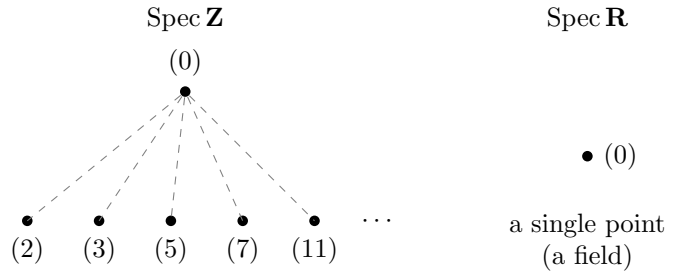
(iii) $\mathfrak{p} \supseteq \bigcup E_i \iff \mathfrak{p} \supseteq E_i$ for all i. So $V(\bigcup E_i) = \bigcap V(E_i)$.

(iv) $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a}, \mathfrak{b}$ gives $V(\mathfrak{a}) \cup V(\mathfrak{b}) \subseteq V(\mathfrak{a} \cap \mathfrak{b}) \subseteq V(\mathfrak{a}\mathfrak{b})$. If $\mathfrak{p} \supseteq \mathfrak{a}\mathfrak{b}$ and $\mathfrak{p} \not\supseteq \mathfrak{a}$, pick $x \in \mathfrak{a} \setminus \mathfrak{p}$; for each $y \in \mathfrak{b}$, $xy \in \mathfrak{p}$, so $y \in \mathfrak{p}$, i.e. $\mathfrak{p} \supseteq \mathfrak{b}$. So $V(\mathfrak{a}\mathfrak{b}) \subseteq V(\mathfrak{a}) \cup V(\mathfrak{b})$. All four sets agree. □

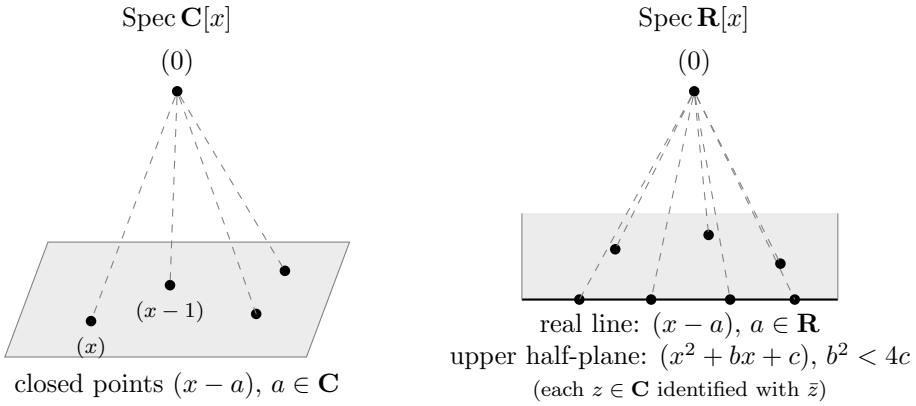
Exercise 16

Problem. Draw pictures of $\operatorname{Spec}(\mathbf{Z})$, $\operatorname{Spec}(\mathbf{R})$, $\operatorname{Spec}(\mathbf{C}[x])$, $\operatorname{Spec}(\mathbf{R}[x])$, $\operatorname{Spec}(\mathbf{Z}[x])$.

Solution. In each picture a line from a point $\mathfrak{q}$ upward to $\mathfrak{p}$ means $\mathfrak{q} \subset \mathfrak{p}$ (so $\mathfrak{p}$ lies in the closure of $\mathfrak{q}$: $\mathfrak{q}$ is "more generic"). Closed sets are $V(E)$; closed points are maximal ideals.



$\operatorname{Spec}\mathbf{Z}$*:* points (0) and (p), p prime. $V(n) = \{(p) : p \mid n\}$ for $n \neq 0$ is finite, so the proper closed sets are the finite sets of closed points; (0) is dense (the generic point). $\operatorname{Spec}\mathbf{R}$*:* one point (0).

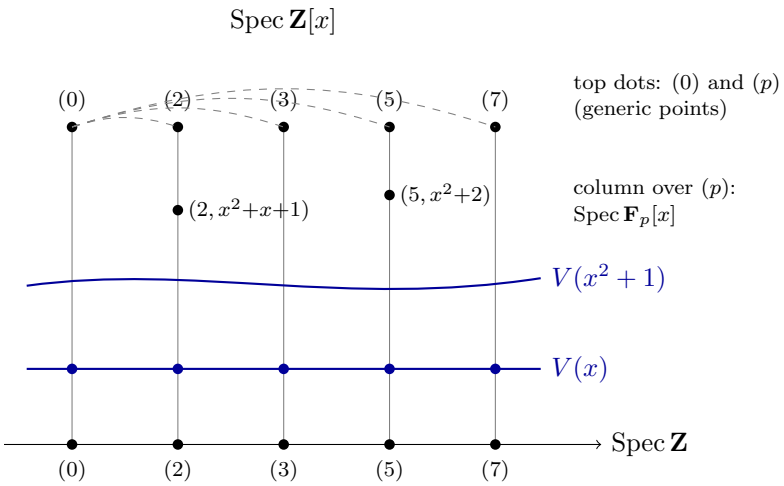


$\operatorname{Spec}\mathbf{C}[x]$*:* $\mathbf{C}[x]$ is a PID, nonzero primes are $(x-a)$ (all maximal), so the space is the complex line $\mathbf{C}$ together with a generic point (0); proper closed sets are finite sets of closed points. $\operatorname{Spec}\mathbf{R}[x]$*:* nonzero primes are $(x-a)$ and (x^2+bx+c) with $b^2 < 4c$. The latter correspond to pairs of conjugate roots $\{z, \bar{z}\}$, so the closed points form $\mathbf{C}/(z \sim \bar{z}) \cong \{\operatorname{Im} z \geq 0\}$: the real axis (for $(x-a)$) and the open upper half-plane. Plus the generic point.

$\operatorname{Spec}\mathbf{Z}[x]$. Let $\mathfrak{p}$ be prime and $\mathfrak{q} = \mathfrak{p} \cap \mathbf{Z}$.

- $\mathfrak{q} = 0$: $\mathfrak{p}$ corresponds to a prime of $\mathbf{Q}[x]$ (localize), so $\mathfrak{p} = 0$ or $\mathfrak{p} = (f)$ with f irreducible and primitive (Gauss's lemma), e.g. $(x), (x^2+1), (2x-1)$.
- $\mathfrak{q} = (p)$: $\mathfrak{p}/(p)$ is a prime of $\mathbf{F}_p[x]$, so $\mathfrak{p} = (p)$ or $\mathfrak{p} = (p, g)$ with g monic and irreducible mod p. These last are exactly the maximal ideals.

Picture: the map $\operatorname{Spec}\mathbf{Z}[x] \to \operatorname{Spec}\mathbf{Z}$ has fibre $\operatorname{Spec}\mathbf{F}_p[x]$ (a line with generic point (p)) over (p) and fibre $\operatorname{Spec}\mathbf{Q}[x]$ over (0). The curves $V(f)$ cut every column.



□

Exercise 17

Problem. For each $f \in A$, let X_f denote the complement of $V(f)$ in $X = \mathrm{Spec}(A)$. The sets X_f are open. Show that they form a basis of open sets for the Zariski topology, and that

(i) $X_f \cap X_g = X_{fg}$;

(ii) $X_f = \emptyset \iff f$ is nilpotent;

(iii) $X_f = X \iff f$ is a unit;

(iv) $X_f = X_g \iff r((f)) = r((g))$;

(v) X is quasi-compact (that is, every open covering of X has a finite subcovering);

(vi) more generally, each X_f is quasi-compact;

(vii) an open subset of X is quasi-compact if and only if it is a finite union of sets X_f.

The sets X_f are called the *basic open sets* of $X = \mathrm{Spec}(A)$. [To prove (v), remark that it is enough to consider a covering of X by basic open sets X_{f_i} $(i \in I)$. Show that the f_i generate the unit ideal and hence that there is an equation of the form $1 = \sum_{i \in J} g_i f_i$ $(g_i \in A)$ where J is some *finite* subset of I. Then the X_{f_i} $(i \in J)$ cover X.]

Solution. Note $X_f = X \setminus V(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$, and $V(f) := V(\{f\}) = V((f))$.

Basis. Every open set is $X \setminus V(E) = X \setminus \bigcap_{f \in E} V(f) = \bigcup_{f \in E} X_f$ (15(iii)). So the X_f form a basis.

(i) $\mathfrak{p} \notin V(f) \cup V(g) \iff f, g \notin \mathfrak{p} \iff fg \notin \mathfrak{p}$ (prime), so $X_f \cap X_g = X_{fg}$.

(ii) $X_f = \emptyset \iff f$ lies in every prime $\iff f \in \mathfrak{N}$ (1.8).

(iii) $X_f = X \iff f$ lies in no prime $\iff f$ lies in no maximal ideal $\iff f$ is a unit (1.5).

(iv) $X_f = X_g \iff V(f) = V(g) \iff r((f)) = r((g))$: since $r(\mathfrak{a}) = \bigcap_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p}$ (1.14), $V(\mathfrak{a}) = V(\mathfrak{b})$ implies $r\mathfrak{a} = r\mathfrak{b}$, and the converse is 15(i).

(v) Every open cover can be refined to a cover by basic open sets X_{f_i} ($i \in I$), and a finite subcover of the refinement gives a finite subcover of the original (pick for each of finitely many X_{f_i} one member of the original cover containing it). So take $X = \bigcup_{i\in I} X_{f_i}$. Then $V(\{f_i\}) = \bigcap V(f_i) = \emptyset$, so the ideal generated by the f_i is (1) (else it lies in a maximal ideal, which is a prime in $V(\{f_i\})$). Hence $1 = \sum_{i\in J} g_i f_i$ with $J \subseteq I$ finite. If a prime $\mathfrak{p}$ were outside $\bigcup_{i\in J} X_{f_i}$, all $f_i \in \mathfrak{p}$ ($i \in J$), giving $1 \in \mathfrak{p}$. So $X = \bigcup_{i\in J} X_{f_i}$.

(vi) As in (v) it suffices to consider $X_f \subseteq \bigcup_{i\in I} X_{g_i}$. Then every prime containing all g_i contains f: $f \in \bigcap_{\mathfrak{p}\supseteq(g_i)} \mathfrak{p} = r((g_i : i \in I))$, so $f^n = \sum_{i\in J} h_i g_i$ for some n and finite J. Then $X_f = X_{f^n} \subseteq \bigcup_{i\in J} X_{g_i}$: a prime $\mathfrak{p}$ with $f \notin \mathfrak{p}$ cannot contain all g_i, $i \in J$ (it would contain f^n).

(vii) ($\Leftarrow$) A finite union of quasi-compact sets is quasi-compact, and X_f is quasi-compact by (vi). ($\Rightarrow$) An open U is a union of basic opens; if U is quasi-compact, finitely many suffice. □

Exercise 18

Problem. For psychological reasons it is sometimes convenient to denote a prime ideal of A by a letter such as x or y when thinking of it as a point of $X = \mathrm{Spec}(A)$. When thinking of x as a prime ideal of A, we denote it by $\mathfrak{p}_x$ (logically, of course, it is the same thing). Show that

(i) the set $\{x\}$ is closed (we say that x is a "closed point") in $\mathrm{Spec}(A) \iff \mathfrak{p}_x$ is maximal;
(ii) $\overline{\{x\}} = V(\mathfrak{p}_x)$;
(iii) $y \in \overline{\{x\}} \iff \mathfrak{p}_x \subseteq \mathfrak{p}_y$;
(iv) X is a T_0-space (this means that if x, y are distinct points of X, then either there is a neighborhood of x which does not contain y, or else there is a neighborhood of y which does not contain x).

Solution. Write $x \leftrightarrow \mathfrak{p}_x$.

(ii) A closed set containing x is $V(E)$ with $E \subseteq \mathfrak{p}_x$, hence $V(E) \supseteq V(\mathfrak{p}_x)$. So the smallest closed set containing x is $V(\mathfrak{p}_x)$: $\overline{\{x\}} = V(\mathfrak{p}_x)$.

(i) By (ii), $\{x\}$ is closed $\iff V(\mathfrak{p}_x) = \{\mathfrak{p}_x\} \iff$ no prime strictly contains $\mathfrak{p}_x \iff \mathfrak{p}_x$ is maximal (if $\mathfrak{p}_x$ is not maximal it lies in a maximal ideal, a different prime; the converse is clear).

(iii) $y \in \overline{\{x\}} = V(\mathfrak{p}_x) \iff \mathfrak{p}_x \subseteq \mathfrak{p}_y$.

(iv) Let $x \neq y$, i.e. $\mathfrak{p}_x \neq \mathfrak{p}_y$. Then $\mathfrak{p}_x \not\subseteq \mathfrak{p}_y$ or $\mathfrak{p}_y \not\subseteq \mathfrak{p}_x$. In the first case $y \notin \overline{\{x\}} = V(\mathfrak{p}_x)$, so $X \setminus V(\mathfrak{p}_x)$ is an open neighbourhood of y not containing x; in the second case symmetrically there is a neighbourhood of x missing y. □

Exercise 19

Problem. A topological space X is said to be *irreducible* if $X \neq \emptyset$ and if every pair of non-empty open sets in X intersect, or equivalently if every non-empty open set is dense in X. Show that $\mathrm{Spec}(A)$ is irreducible if and only if the nilradical of A is a prime ideal.

Solution. Every non-empty open set contains a non-empty basic open set, so X is irreducible iff $X \neq \emptyset$ and any two non-empty basic opens meet. Now $X \neq \emptyset \iff A \neq 0 \iff 1 \notin \mathfrak{N} \iff$

$\mathfrak{N} \neq (1)$. By 17(ii), $X_f \neq \emptyset \iff f \notin \mathfrak{N}$, and $X_f \cap X_g = X_{fg}$ (17(i)). So the second condition reads: $f, g \notin \mathfrak{N} \Rightarrow fg \notin \mathfrak{N}$, i.e. $\mathfrak{N}$ is prime (together with $\mathfrak{N} \neq (1)$). □

Exercise 20

Problem. Let X be a topological space.

(i) If Y is an irreducible (Exercise 19) subspace of X, then the closure $\bar{Y}$ of Y in X is irreducible.

(ii) Every irreducible subspace of X is contained in a maximal irreducible subspace.

(iii) The maximal irreducible subspaces of X are closed and cover X. They are called the *irreducible components* of X. What are the irreducible components of a Hausdorff space?

(iv) If A is a ring and $X = \mathrm{Spec}(A)$, then the irreducible components of X are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal of A (Exercise 8).

Solution. (i) Let Y be irreducible and U, V non-empty open subsets of $\bar{Y}$. Pick a point of U; as it lies in the closure of Y and U is open in $\bar{Y}$ (so $U = \bar{Y} \cap U'$, U' open in X), U' meets Y; so $U \cap Y \neq \emptyset$, similarly $V \cap Y \neq \emptyset$. These are non-empty open subsets of Y, hence meet, so $U \cap V \neq \emptyset$. Also $\bar{Y} \neq \emptyset$.

(ii) Let Y be irreducible and apply Zorn to the set of irreducible subspaces containing Y, ordered by inclusion. If (Y_α) is a chain, $Z = \bigcup Y_\alpha$ is irreducible: if U, V are non-empty open in Z, choose $u \in U \cap Y_\alpha$, $v \in V \cap Y_\beta$ with (say) $Y_\alpha \subseteq Y_\beta$; then $U \cap Y_\beta, V \cap Y_\beta$ are non-empty and open in Y_β so they meet. Hence maximal irreducible subspaces exist above every irreducible subspace.

(iii) If Y is maximal irreducible then $\bar{Y}$ is irreducible by (i) and contains Y, so $Y = \bar{Y}$ is closed. Each point $\{x\}$ is irreducible, so by (ii) lies in a maximal irreducible subspace; these cover X.

Hausdorff spaces. If Y is irreducible with two distinct points x, y, take disjoint open neighbourhoods $U \ni x$, $V \ni y$ in X; then $U \cap Y$, $V \cap Y$ are disjoint non-empty open sets of Y, contradiction. So irreducible subspaces are single points: the irreducible components of a Hausdorff space are its points.

(iv) Let $X = \mathrm{Spec}\, A$. For a closed set $Y = V(\mathfrak{a})$ put $I(Y) = \bigcap_{\mathfrak{q} \in Y} \mathfrak{q}$; then $Y = V(I(Y))$ (as $V(\mathfrak{a}) = V(r\mathfrak{a})$ and $I(Y) = r\mathfrak{a}$). *Claim: a closed $Y \neq \emptyset$ is irreducible $\iff$ $I(Y)$ is prime.* ($\Rightarrow$) $I(Y) \neq (1)$. If $xy \in I(Y)$, then every $\mathfrak{q} \in Y$ contains x or y, so $Y = (Y \cap V(x)) \cup (Y \cap V(y))$. If both pieces were proper, their complements in Y would be non-empty disjoint open subsets of Y, contradicting irreducibility. So $Y \subseteq V(x)$ or $V(y)$, i.e. $x \in I(Y)$ or $y \in I(Y)$. ($\Leftarrow$) Let $\mathfrak{p} = I(Y)$ be prime, so $Y = V(\mathfrak{p})$. If $Y = Y_1 \cup Y_2$ with Y_i closed, then $\mathfrak{p} = I(Y_1) \cap I(Y_2) \supseteq I(Y_1)I(Y_2)$, so (primality) $\mathfrak{p} \supseteq I(Y_1)$, say; then $Y = V(\mathfrak{p}) \subseteq V(I(Y_1)) = Y_1$. So Y is not a union of two proper closed subsets, i.e. irreducible.

Hence the closed irreducible subsets are exactly the $V(\mathfrak{p})$, $\mathfrak{p}$ prime, and $V(\mathfrak{p}) \subseteq V(\mathfrak{q}) \iff \mathfrak{q} \subseteq \mathfrak{p}$. The maximal ones are the $V(\mathfrak{p})$ with $\mathfrak{p}$ a minimal prime. By (iii) maximal irreducible subspaces are closed, so these are exactly the irreducible components. □

Exercise 21

Problem. Let $\phi : A \to B$ be a ring homomorphism. Let $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. If $\mathfrak{q} \in Y$, then $\phi^{-1}(\mathfrak{q})$ is a prime ideal of A, i.e. a point of X. Hence ϕ induces a mapping $\phi^* : Y \to X$. Show that

(i) If $f \in A$ then $\phi^{*-1}(X_f) = Y_{\phi(f)}$, and hence that ϕ^* is continuous.

(ii) If $\mathfrak{a}$ is an ideal of A, then $\phi^{*-1}(V(\mathfrak{a})) = V(\mathfrak{a}^{\mathrm{e}})$.

(iii) If $\mathfrak{b}$ is an ideal of B, then $\overline{\phi^*(V(\mathfrak{b}))} = V(\mathfrak{b}^{\mathrm{c}})$.

(iv) If ϕ is surjective, then ϕ^* is a homeomorphism of Y onto the closed subset $V(\mathrm{Ker}(\phi))$ of X. (In particular, $\mathrm{Spec}(A)$ and $\mathrm{Spec}(A/\mathfrak{N})$, where $\mathfrak{N}$ is the nilradical of A, are naturally homeomorphic.)

(v) If ϕ is injective, then $\phi^*(Y)$ is dense in X. More precisely, $\phi^*(Y)$ is dense in $X \iff \mathrm{Ker}(\phi) \subseteq \mathfrak{N}$.

(vi) Let $\psi : B \to C$ be another ring homomorphism. Then $(\psi \circ \phi)^* = \phi^* \circ \psi^*$.

(vii) Let A be an integral domain with just one non-zero prime ideal $\mathfrak{p}$, and let K be the field of fractions of A. Let $B = (A/\mathfrak{p}) \times K$. Define $\phi : A \to B$ by $\phi(x) = (\bar{x}, x)$, where $\bar{x}$ is the image of x in $A/\mathfrak{p}$. Show that ϕ^* is bijective but not a homeomorphism.

Solution. Let $Y = \mathrm{Spec}\, B$, $\phi^*(\mathfrak{q}) = \phi^{-1}(\mathfrak{q})$.

(i) $\mathfrak{q} \in \phi^{*-1}(X_f) \iff f \notin \phi^{-1}(\mathfrak{q}) \iff \phi(f) \notin \mathfrak{q} \iff \mathfrak{q} \in Y_{\phi(f)}$. Preimages of basic opens are open, so ϕ^* is continuous.

(ii) $\mathfrak{q} \in \phi^{*-1}(V(\mathfrak{a})) \iff \phi^{-1}(\mathfrak{q}) \supseteq \mathfrak{a} \iff \mathfrak{q} \supseteq \phi(\mathfrak{a}) \iff \mathfrak{q} \supseteq \mathfrak{a}^{\mathrm{e}}$.

(iii) The closure of $S \subseteq X$ is $V\big(\bigcap_{\mathfrak{p} \in S} \mathfrak{p}\big)$ (the smallest $V(E)$ containing S has $E \subseteq \bigcap_S \mathfrak{p}$). For $S = \phi^*(V(\mathfrak{b})) = \{\phi^{-1}(\mathfrak{q}) : \mathfrak{q} \supseteq \mathfrak{b}\}$:

$$\bigcap_{\mathfrak{q} \supseteq \mathfrak{b}} \phi^{-1}(\mathfrak{q}) = \phi^{-1}\Big(\bigcap_{\mathfrak{q} \supseteq \mathfrak{b}} \mathfrak{q}\Big) = \phi^{-1}(r\mathfrak{b}) = r(\mathfrak{b}^{\mathrm{c}})$$

(by 1.14 and Ex. 1.18). Hence $\overline{\phi^*(V(\mathfrak{b}))} = V(r(\mathfrak{b}^{\mathrm{c}})) = V(\mathfrak{b}^{\mathrm{c}})$ by 15(i).

(iv) Let ϕ be surjective. ϕ^* maps into $V(\mathrm{Ker}\,\phi)$ since $\phi^{-1}(\mathfrak{q}) \supseteq \phi^{-1}(0)$. It is injective: $\phi(\phi^{-1}(\mathfrak{q})) = \mathfrak{q}$ by surjectivity, so $\phi^{-1}(\mathfrak{q}) = \phi^{-1}(\mathfrak{q}') \Rightarrow \mathfrak{q} = \mathfrak{q}'$. It is onto $V(\mathrm{Ker}\,\phi)$: if $\mathfrak{p} \supseteq \mathrm{Ker}\,\phi$ is prime then $\phi(\mathfrak{p})$ is a prime of B and $\phi^{-1}(\phi(\mathfrak{p})) = \mathfrak{p} + \mathrm{Ker}\,\phi = \mathfrak{p}$. It is closed: for $\mathfrak{b} \subseteq B$, $\phi^*(V(\mathfrak{b})) = V(\mathfrak{b}^{\mathrm{c}})$ (given $\mathfrak{p} \supseteq \mathfrak{b}^{\mathrm{c}}$ then $\phi(\mathfrak{p}) \supseteq \phi(\phi^{-1}\mathfrak{b}) = \mathfrak{b}$ and $\phi^*(\phi(\mathfrak{p})) = \mathfrak{p}$; the other inclusion from (iii)). A continuous closed bijection onto $V(\mathrm{Ker}\,\phi)$ is a homeomorphism. For $A \to A/\mathfrak{N}$ we have $V(\mathfrak{N}) = X$, so $\mathrm{Spec}(A/\mathfrak{N}) \cong \mathrm{Spec}\, A$.

(v) By (iii) with $\mathfrak{b} = 0$: $\overline{\phi^*(Y)} = V(\mathrm{Ker}\,\phi)$. This equals X iff every prime contains $\mathrm{Ker}\,\phi$ iff $\mathrm{Ker}\,\phi \subseteq \mathfrak{N}$. In particular if ϕ is injective, $\phi^*(Y)$ is dense.

(vi) $(\psi\phi)^{-1}(\mathfrak{r}) = \phi^{-1}(\psi^{-1}(\mathfrak{r}))$, i.e. $(\psi \circ \phi)^* = \phi^* \circ \psi^*$.

(vii) Let A be a domain with exactly one non-zero prime $\mathfrak{p}$, $K = \mathrm{Frac}\, A$, $B = (A/\mathfrak{p}) \times K$, $\phi(x) = (\bar{x}, x)$ (an injective ring homomorphism). Since A has a non-zero prime it is not a field, so it has a maximal ideal, which is a non-zero prime, hence $\mathfrak{p}$ is maximal and $A/\mathfrak{p}$ is a field. Thus B is a product of two fields and $\mathrm{Spec}\, B = \{\mathfrak{q}_1, \mathfrak{q}_2\}$ with $\mathfrak{q}_1 = 0 \times K$, $\mathfrak{q}_2 = (A/\mathfrak{p}) \times 0$; both are closed points, so $\mathrm{Spec}\, B$ is discrete. We have $\phi^*(\mathfrak{q}_1) = \{x : \bar{x} = 0\} = \mathfrak{p}$ and $\phi^*(\mathfrak{q}_2) = \{x : x = 0\} = 0$. So ϕ^* is a bijection onto $\mathrm{Spec}\, A = \{0, \mathfrak{p}\}$. In $\mathrm{Spec}\, A$ the closed sets are $\emptyset, \{\mathfrak{p}\} = V(\mathfrak{p}), X$ (the

point 0 is dense), so $\{0\}$ is open but $\{\mathfrak{p}\}$ is not open; yet $\{\mathfrak{q}_1\}$ is open in $\operatorname{Spec} B$ and maps to $\{\mathfrak{p}\}$. So ϕ^* is not open, hence not a homeomorphism. □

Exercise 22

Problem. Let $A = \prod_{i=1}^n A_i$ be the direct product of rings A_i. Show that $\operatorname{Spec}(A)$ is the disjoint union of open (and closed) subspaces X_i, where X_i is canonically homeomorphic with $\operatorname{Spec}(A_i)$.

Conversely, let A be any ring. Show that the following statements are equivalent:

(i) $X = \operatorname{Spec}(A)$ is disconnected.

(ii) $A \cong A_1 \times A_2$ where neither of the rings A_1, A_2 is the zero ring.

(iii) A contains an idempotent $\neq 0, 1$.

In particular, the spectrum of a local ring is always connected (Exercise 12).

Solution. Part 1. Let $A = \prod_{i=1}^n A_i$ and $e_i = (0, \ldots, 1, \ldots, 0)$. These are idempotents with $e_i e_j = 0$ $(i \neq j)$, $\sum e_i = 1$. For a prime $\mathfrak{p}$: $e_i(1 - e_i) = 0$ so $e_i \in \mathfrak{p}$ or $1 - e_i \in \mathfrak{p}$, not both. Since $e_i e_j = 0 \in \mathfrak{p}$ at most one $e_i \notin \mathfrak{p}$, and not all lie in $\mathfrak{p}$ (as $\sum e_i = 1$). So $X = \bigsqcup_i X_{e_i}$ (disjoint union of open sets), hence each X_{e_i} is also closed. Moreover $X_{e_i} = V(1 - e_i)$ (by the above dichotomy). Let $p_i : A \to A_i$ be the projection, surjective with kernel $(1 - e_i)$. By 21(iv), $\operatorname{Spec} A_i \to V(\operatorname{Ker} p_i) = V(1 - e_i) = X_{e_i}$ is a homeomorphism.

Part 2. For a ring A we show (i)⇒(iii)⇒(ii)⇒(i).

(i)⇒(iii). Let $X = V(\mathfrak{a}) \sqcup V(\mathfrak{b})$ with both non-empty (disconnected means a disjoint union of two non-empty closed, equivalently open, sets). From $V(\mathfrak{a}) \cap V(\mathfrak{b}) = V(\mathfrak{a} + \mathfrak{b}) = \emptyset$ we get $\mathfrak{a} + \mathfrak{b} = (1)$ (else a maximal ideal contains both). From $V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = X$ we get $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{N}$. Write $1 = x + y$ with $x \in \mathfrak{a}$, $y \in \mathfrak{b}$; then $xy \in \mathfrak{N}$, say $(xy)^n = 0$. Expanding $1 = (x + y)^{2n} = \sum_i \binom{2n}{i} x^i y^{2n-i}$, each term has $i \geq n$ (so lies in (x^n)) or $2n - i > n$ (so lies in (y^n)); thus $1 \in (x^n, y^n)$: write $1 = e + f$, $e \in (x^n) \subseteq \mathfrak{a}$, $f \in (y^n) \subseteq \mathfrak{b}$. Then $ef \in (x^n y^n) = 0$, so $e(1 - e) = ef = 0$: e is idempotent. If $e = 0$ then $1 = f \in \mathfrak{b}$, $V(\mathfrak{b}) = \emptyset$, contradiction; if $e = 1$ then $1 \in \mathfrak{a}$, $V(\mathfrak{a}) = \emptyset$, contradiction. So $e \neq 0, 1$.

(iii)⇒(ii). For an idempotent $e \neq 0, 1$, the map $A \to Ae \times A(1 - e)$, $x \mapsto (xe, x(1 - e))$ is a ring isomorphism (injective: $xe = 0 = x(1 - e) \Rightarrow x = 0$; surjective: $ae + b(1 - e) \mapsto (ae, b(1 - e))$), where Ae has identity $e \neq 0$ and $A(1 - e)$ has identity $1 - e \neq 0$; neither factor is the zero ring.

(ii)⇒(i). By Part 1, $\operatorname{Spec}(A_1 \times A_2) = X_1 \sqcup X_2$ with $X_i \cong \operatorname{Spec} A_i$ non-empty (since $A_i \neq 0$), open and closed.

Local rings. A local ring has no idempotents $\neq 0, 1$ (Ex. 12), so (iii) fails and $\operatorname{Spec} A$ is connected. □

Exercise 23

Problem. Let A be a Boolean ring (Exercise 11), and let $X = \operatorname{Spec}(A)$.

(i) For each $f \in A$, the set X_f (Exercise 17) is both open and closed in X.

(ii) Let $f_1, \ldots, f_n \in A$. Show that $X_{f_1} \cup \cdots \cup X_{f_n} = X_f$ for some $f \in A$.

(iii) The sets X_f are the only subsets of X which are both open and closed. [Let $Y \subseteq X$ be both open and closed. Since Y is open, it is a union of basic open sets X_f. Since Y is closed and X is quasi-compact (Exercise 17), Y is quasi-compact. Hence Y is a finite union of basic open sets; now use (ii) above.]

(iv) X is a compact Hausdorff space.

Solution. Let A be Boolean, so $f(1-f) = 0$ for all f.

(i) Each prime contains exactly one of $f, 1-f$; so $X_f = \{\mathfrak{p} : f \notin \mathfrak{p}\} = \{\mathfrak{p} : 1-f \in \mathfrak{p}\} = V(1-f)$ is closed. It is open by definition.

(ii) By Ex. 11(iii), $(f_1, \ldots, f_n) = (f)$ for some f. Then $V(f) = V(\{f_1, \ldots, f_n\}) = \bigcap V(f_i)$, so $X_f = \bigcup X_{f_i}$.

(iii) Each X_f is clopen. Conversely let Y be clopen. As Y is open it is a union of basic open sets; as Y is closed in the quasi-compact space X (17(v)), Y is quasi-compact; hence $Y = X_{f_1} \cup \cdots \cup X_{f_n} = X_f$ by (ii).

(iv) X is quasi-compact (17(v)). Hausdorff: let $\mathfrak{p} \neq \mathfrak{q}$ be primes; they are maximal (Ex. 11(ii)), so $\mathfrak{p} \not\subseteq \mathfrak{q}$, and there is $f \in \mathfrak{p} \setminus \mathfrak{q}$. Then $\mathfrak{q} \in X_f$, and $1-f \notin \mathfrak{p}$ (else $1 \in \mathfrak{p}$) so $\mathfrak{p} \in X_{1-f}$, and $X_f \cap X_{1-f} = X_{f(1-f)} = X_0 = \emptyset$. So X is compact Hausdorff. □

Exercise 24

Problem. Let L be a lattice, in which the sup and inf of two elements a, b are denoted by $a \vee b$ and $a \wedge b$ respectively. L is a *Boolean lattice* (or *Boolean algebra*) if

(i) L has a least element and a greatest element (denoted by 0, 1 respectively).

(ii) Each of $\vee, \wedge$ is distributive over the other.

(iii) Each $a \in L$ has a unique "complement" $a' \in L$ such that $a \vee a' = 1$ and $a \wedge a' = 0$.

(For example, the set of all subsets of a set, ordered by inclusion, is a Boolean lattice.)

Let L be a Boolean lattice. Define addition and multiplication in L by the rules

$$a + b = (a \wedge b') \vee (a' \wedge b), \qquad ab = a \wedge b.$$

Verify that in this way L becomes a Boolean ring, say $A(L)$.

Conversely, starting from a Boolean ring A, define an ordering on A as follows: $a \leq b$ means that $a = ab$. Show that, with respect to this ordering, A is a Boolean lattice. [The sup and inf are given by $a \vee b = a + b + ab$ and $a \wedge b = ab$, and the complement by $a' = 1 - a$.] In this way we obtain a one-to-one correspondence between (isomorphism classes of) Boolean rings and (isomorphism classes of) Boolean lattices.

Solution. Lattice facts. In a Boolean lattice L: $a \wedge 1 = a$, $a \vee 0 = a$, $a \wedge 0 = 0$, $a \vee 1 = 1$; $a'' = a$ (since a is a complement of a' and complements are unique); and De Morgan: $(a \vee b)' = a' \wedge b'$, because $(a \vee b) \vee (a' \wedge b') = (a \vee b \vee a') \wedge (a \vee b \vee b') = 1$ and $(a \vee b) \wedge a' \wedge b' = 0$ (distributivity), and by uniqueness; dually $(a \wedge b)' = a' \vee b'$.

$L \mapsto A(L)$ is a Boolean ring. Define $a + b = (a \wedge b') \vee (a' \wedge b)$, $ab = a \wedge b$.

- $+$ is commutative (the formula is symmetric); $a + 0 = (a \wedge 1) \vee (a' \wedge 0) = a$; $a + a = (a \wedge a') \vee (a' \wedge a) = 0$, so $-a = a$.
- Associativity. If $s = a + b$ then $s' = (a' \vee b) \wedge (a \vee b') = (a \wedge b) \vee (a' \wedge b')$ (De Morgan and distributivity). So

$$(a + b) + c = (s \wedge c') \vee (s' \wedge c) = (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (a' \wedge b' \wedge c) \vee (a \wedge b \wedge c),$$

  which is symmetric in a, b, c. Hence $(a + b) + c = (b + c) + a = a + (b + c)$.
- $\wedge$ is associative, commutative, with identity 1.
- Distributivity: $a(b+c) = (a \wedge b \wedge c') \vee (a \wedge b' \wedge c)$, while $ab + ac = \big((a \wedge b) \wedge (a' \vee c')\big) \vee \big((a' \vee b') \wedge a \wedge c\big) = (a \wedge b \wedge c') \vee (a \wedge b' \wedge c)$ (terms containing $a \wedge a'$ vanish).
- $a \cdot a = a \wedge a = a$: Boolean.

$A \mapsto L(A)$ **is a Boolean lattice.** Let A be Boolean (so $2x = 0$) and define $a \leq b \iff a = ab$.

- Partial order: $a = a \cdot a$; $a = ab, b = ba \Rightarrow a = b$; $a = ab, b = bc \Rightarrow a = abc = ac$.
- $a \wedge b = ab$: $ab \leq a, b$; and $c \leq a, c \leq b$ give $c \cdot ab = (ca)b = cb = c$. $a \vee b = a + b + ab$: $a(a+b+ab) = a+2ab = a$, so $a \leq a+b+ab$, same for b; if $a, b \leq c$ then $(a+b+ab)c = a+b+ab$.
- Least element 0 ($0 = 0 \cdot a$), greatest element 1 ($a = a \cdot 1$).
- $a(b \vee c) = ab + ac + abc = (a \wedge b) \vee (a \wedge c)$; and $(a \vee b) \wedge (a \vee c) = (a + b + ab)(a + c + ac) = a + bc + abc = a \vee (b \wedge c)$ (collecting the nine products and using $2x = 0$).
- Complement $a' = 1 - a = 1 + a$: $a \wedge a' = a + a^2 = 0$, $a \vee a' = a + (1 + a) + 0 = 1$, and if $ac = 0$, $a + c + ac = 1$ then $c = 1 + a$: unique.

The two constructions are inverse. Starting from A, form $L(A)$ and then $A(L(A))$: its product is $a \wedge b = ab$, and its sum is $(a \wedge b') \vee (a' \wedge b) = x \vee y$ with $x = a(1 + b) = a + ab$, $y = b + ab$; since $xy = ab + ab + ab + ab = 0$, $x \vee y = x + y = a + b + 2ab = a + b$. So we recover A. Starting from L, form $A(L)$ and then $L(A(L))$: $a \leq b \iff a = a \wedge b$ is the original order of L, and $\vee, \wedge$ are determined by the order. So we recover L. Hence the two constructions give a bijection between isomorphism classes of Boolean rings and Boolean lattices. $\square$

Exercise 25

> **Problem.** From the last two exercises deduce Stone's theorem, that every Boolean lattice is isomorphic to the lattice of open-and-closed subsets of some compact Hausdorff topological space.

Solution. Let L be a Boolean lattice, $A = A(L)$ its Boolean ring (Ex. 24), $X = \operatorname{Spec} A$, a compact Hausdorff space (Ex. 23). By 23(i),(iii) the clopen subsets of X are exactly the X_f, $f \in A$. Define $\theta : A \to \{\text{clopen sets}\}$, $f \mapsto X_f$; it is onto.

Injective. If $X_f = X_g$ then $r((f)) = r((g))$ (17(iv)). In a Boolean ring $g^n = g$, so $r(\mathfrak{a}) = \mathfrak{a}$ for every ideal $\mathfrak{a}$; hence $(f) = (g)$, i.e. $f = gu$, $g = fv$, so $fg = gug = gu = f$ and likewise $fg = g$: $f = g$.

Order. $f \leq g \iff f = fg \iff X_f = X_{fg} = X_f \cap X_g \iff X_f \subseteq X_g$ (using 17(i) and injectivity of θ for $\Leftarrow$).

A bijection preserving the order in both directions is an isomorphism of lattices (sup and inf are determined by the order). Since the order of L is the order of $L(A(L))$, $L \cong$ the lattice of open-and-closed subsets of the compact Hausdorff space X under $\cup, \cap$. $\square$

Exercise 26

Problem. Let A be a ring. The subspace of $\mathrm{Spec}(A)$ consisting of the maximal ideals of A, with the induced topology, is called the *maximal spectrum* of A and is denoted by $\mathrm{Max}(A)$. For arbitrary commutative rings it does not have the nice functorial properties of $\mathrm{Spec}(A)$ (see Exercise 21), because the inverse image of a maximal ideal under a ring homomorphism need not be maximal.

Let X be a compact Hausdorff space and let $C(X)$ denote the ring of all real-valued continuous functions on X (add and multiply functions by adding and multiplying their values). For each $x \in X$, let $\mathfrak{m}_x$ be the set of all $f \in C(X)$ such that $f(x) = 0$. The ideal $\mathfrak{m}_x$ is maximal, because it is the kernel of the (surjective) homomorphism $C(X) \to \mathbf{R}$ which takes f to $f(x)$. If $\tilde{X}$ denotes $\mathrm{Max}(C(X))$, we have therefore defined a mapping $\mu : X \to \tilde{X}$, namely $x \mapsto \mathfrak{m}_x$.

We shall show that μ is a homeomorphism of X onto $\tilde{X}$.

(i) Let $\mathfrak{m}$ be any maximal ideal of $C(X)$, and let $V = V(\mathfrak{m})$ be the set of common zeros of the functions in $\mathfrak{m}$: that is, $V = \{x \in X : f(x) = 0 \text{ for all } f \in \mathfrak{m}\}$. Suppose that V is empty. Then for each $x \in X$ there exists $f_x \in \mathfrak{m}$ such that $f_x(x) \neq 0$. Since f_x is continuous, there is an open neighborhood U_x of x in X on which f_x does not vanish. By compactness a finite number of the neighborhoods, say $U_{x_1}, \ldots, U_{x_n}$, cover X. Let $f = f_{x_1}^2 + \cdots + f_{x_n}^2$. Then f does not vanish at any point of X, hence is a unit in $C(X)$. But this contradicts $f \in \mathfrak{m}$. Hence V is not empty. Let x be a point of V. Then $\mathfrak{m} \subseteq \mathfrak{m}_x$, hence $\mathfrak{m} = \mathfrak{m}_x$ because $\mathfrak{m}$ is maximal. Hence μ is surjective.

(ii) By Urysohn's lemma (this is the only non-trivial fact required in the argument) the continuous functions separate the points of X. Hence $x \neq y \Rightarrow \mathfrak{m}_x \neq \mathfrak{m}_y$, and therefore μ is injective.

(iii) Let $f \in C(X)$; let $U_f = \{x \in X : f(x) \neq 0\}$ and let $\tilde{U}_f = \{\mathfrak{m} \in \tilde{X} : f \notin \mathfrak{m}\}$. Show that $\mu(U_f) = \tilde{U}_f$. The open sets U_f (resp. $\tilde{U}_f$) form a basis of the topology of X (resp. $\tilde{X}$) and therefore μ is a homeomorphism.

Thus X can be reconstructed from the ring of functions $C(X)$.

Solution. Parts (i) and (ii) are proved in the text (compactness gives a point of $V(\mathfrak{m})$, hence μ is surjective; Urysohn's lemma gives injectivity: for $x \neq y$ there is a continuous f with $f(x) = 0$, $f(y) = 1$, so $f \in \mathfrak{m}_x \setminus \mathfrak{m}_y$). So $\mu : X \to \tilde{X} = \mathrm{Max}\, C(X)$, $x \mapsto \mathfrak{m}_x$, is a bijection. It remains to do (iii).

(iii) For $f \in C(X)$: $x \in U_f \iff f(x) \neq 0 \iff f \notin \mathfrak{m}_x \iff \mu(x) \in \tilde{U}_f$. As μ is bijective, $\mu(U_f) = \tilde{U}_f$.

Bases. The U_f are open (continuity). They form a basis: let $W \subseteq X$ be open and $x \in W$. X is compact Hausdorff hence normal, so Urysohn's lemma gives a continuous $g : X \to [0,1]$ with $g(x) = 1$ and $g = 0$ on the closed set $X \setminus W$. Then $x \in U_g \subseteq W$. The $\tilde{U}_f = \tilde{X} \cap X_f$ form a basis of the subspace topology of $\tilde{X} \subseteq \operatorname{Spec} C(X)$ (the X_f are a basis of Spec).

Thus μ is a bijection carrying a basis of X onto a basis of $\tilde{X}$, so μ and μ^{-1} are open maps, i.e. μ is a homeomorphism. □

Exercise 27

> **Problem.** (*Affine algebraic varieties.*) Let k be an algebraically closed field and let $f_\alpha(t_1, \ldots, t_n) = 0$ be a set of polynomial equations in n variables with coefficients in k. The set X of all points $x = (x_1, \ldots, x_n) \in k^n$ which satisfy these equations is an *affine algebraic variety.*
>
> Consider the set of all polynomials $g \in k[t_1, \ldots, t_n]$ with the property that $g(x) = 0$ for all $x \in X$. This set is an ideal $I(X)$ in the polynomial ring, and is called the *ideal of the variety* X. The quotient ring $P(X) = k[t_1, \ldots, t_n]/I(X)$ is the ring of polynomial functions on X, because two polynomials g, h define the same polynomial function on X if and only if $g - h$ vanishes at every point of X, that is, if and only if $g - h \in I(X)$.
>
> Let ξ_i $(1 \leq i \leq n)$ be the image of t_i in $P(X)$. The ξ_i are the *coordinate functions* on X: if $x \in X$, then $\xi_i(x)$ is the ith coordinate of x. $P(X)$ is generated as a k-algebra by the coordinate functions, and is called the *coordinate ring* (or *affine algebra*) of X.
>
> As in Exercise 26, for each $x \in X$ let $\mathfrak{m}_x$ be the ideal of all $f \in P(X)$ such that $f(x) = 0$; it is a maximal ideal of $P(X)$. Hence, if $\tilde{X} = \operatorname{Max}(P(X))$, we have defined a mapping $\mu : X \to \tilde{X}$, namely $x \mapsto \mathfrak{m}_x$. It is easy to show that μ is injective: if $x \neq y$, we must have $x_i \neq y_i$ for some i, and hence $\xi_i - x_i$ is in $\mathfrak{m}_x$ but not in $\mathfrak{m}_y$, so that $\mathfrak{m}_x \neq \mathfrak{m}_y$. What is less obvious (but still true) is that μ is *surjective.* This is one form of Hilbert's Nullstellensatz (see Chapter 7).

Solution. As printed, Exercise 27 sets up notation: $k = \bar{k}$, $X \subseteq k^n$ an affine variety, $I(X)$ its ideal, $P(X) = k[t_1, \ldots, t_n]/I(X)$, and the map $\mu : X \to \tilde{X} = \operatorname{Max} P(X)$, $x \mapsto \mathfrak{m}_x$, and then *asserts* that μ is injective and ("less obvious but still true") surjective. There is no explicit task, but for completeness the assertions:

$\mathfrak{m}_x$ *is maximal:* it is the kernel of the surjective evaluation $P(X) \to k$, $f \mapsto f(x)$ (surjective since $k \subseteq P(X)$).

μ *injective:* if $x \neq y$ then $x_i \neq y_i$ for some i and the coordinate function $\xi_i - x_i$ lies in $\mathfrak{m}_x \setminus \mathfrak{m}_y$.

μ *surjective* (this is the weak Nullstellensatz): a maximal ideal $\mathfrak{m} \subset P(X)$ pulls back to a maximal ideal $M \subset k[t_1, \ldots, t_n]$ containing $I(X)$; since k is algebraically closed, $M = (t_1 - a_1, \ldots, t_n - a_n)$ for some $a \in k^n$. Every defining equation f_α of X lies in $I(X) \subseteq M$, hence $f_\alpha(a) = 0$, so $a \in X$ and $\mathfrak{m} = \mathfrak{m}_a$. (This uses the Nullstellensatz, proved in Chapter 7.) □

Exercise 28

> **Problem.** Let $f_1, \ldots, f_m$ be elements of $k[t_1, \ldots, t_n]$. They determine a *polynomial mapping* $\phi : k^n \to k^m$: if $x \in k^n$, the coordinates of $\phi(x)$ are $f_1(x), \ldots, f_m(x)$.
>
> Let X, Y be affine algebraic varieties in k^n, k^m respectively. A mapping $\phi : X \to Y$ is said to be *regular* if ϕ is the restriction to X of a polynomial mapping from k^n to k^m.

If η is a polynomial function on Y, then $\eta \circ \phi$ is a polynomial function on X. Hence ϕ induces a k-algebra homomorphism $P(Y) \to P(X)$, namely $\eta \mapsto \eta \circ \phi$. Show that in this way we obtain a one-to-one correspondence between the regular mappings $X \to Y$ and the k-algebra homomorphisms $P(Y) \to P(X)$.

Solution. Let $X \subseteq k^n$, $Y \subseteq k^m$, and $\xi_1, \dots, \xi_m$ the coordinate functions on Y; they generate $P(Y)$ as a k-algebra. Given a regular map $\phi = (f_1, \dots, f_m) : X \to Y$ ($f_j \in k[t_1, \dots, t_n]$), define $\phi^\sharp : P(Y) \to P(X)$, $\eta \mapsto \eta \circ \phi$. It is well defined: if $\eta = h|_Y$ with $h \in k[s_1, \dots, s_m]$, then $\eta \circ \phi = h(f_1, \dots, f_m)|_X \in P(X)$. It is clearly a k-algebra homomorphism. We show

$$\Phi : \{\text{regular maps } X \to Y\} \longrightarrow \mathrm{Hom}_{k\text{-alg}}(P(Y), P(X)), \qquad \phi \mapsto \phi^\sharp$$

is bijective.

Injective. $\phi^\sharp(\xi_j) = \xi_j \circ \phi = \phi_j$, the j-th coordinate of ϕ. If $\phi^\sharp = \psi^\sharp$ then $\phi_j = \psi_j$ as functions on X for all j, so $\phi = \psi$.

Surjective. Let $\theta : P(Y) \to P(X)$ be a k-algebra homomorphism. Choose polynomials $f_j \in k[t_1, \dots, t_n]$ whose restrictions to X represent $\theta(\xi_j)$, and let $\phi = (f_1, \dots, f_m) : k^n \to k^m$. We check $\phi(X) \subseteq Y$. Now Y is the common zero locus of $I(Y)$ (it contains the defining equations of Y and every element of $I(Y)$ vanishes on Y). Let $h \in I(Y) \subseteq k[s_1, \dots, s_m]$; then $h(\xi_1, \dots, \xi_m) = 0$ in $P(Y)$, so applying the k-algebra homomorphism θ:

$$0 = \theta\big(h(\xi_1, \dots, \xi_m)\big) = h\big(\theta(\xi_1), \dots, \theta(\xi_m)\big) = h(f_1, \dots, f_m)|_X.$$

So $h(\phi(x)) = 0$ for all $x \in X$, $h \in I(Y)$; hence $\phi(x) \in Y$. Thus $\phi : X \to Y$ is regular, and $\phi^\sharp(\xi_j) = f_j|_X = \theta(\xi_j)$. The homomorphisms $\phi^\sharp$ and θ agree on the generators ξ_j of $P(Y)$, hence are equal. $\square$

Notes

The textbook PDF is a scan without a text layer; I read the exercise statements from page images. In Ex. 1.18 the quotient inequalities are printed as $(\mathfrak{a}_1 : \mathfrak{a}_2)^{\mathrm{e}} \subseteq (\mathfrak{a}_1^{\mathrm{e}} : \mathfrak{a}_2^{\mathrm{e}})$ and $(\mathfrak{b}_1 : \mathfrak{b}_2)^{\mathrm{c}} \subseteq (\mathfrak{b}_1^{\mathrm{c}} : \mathfrak{b}_2^{\mathrm{c}})$, as stated above. In Ex. 21(v) I proved the general equivalence $\phi^*(Y)$ dense $\iff \operatorname{Ker}\phi \subseteq \mathfrak{N}$, which contains the injective case.